DRIVE RATIO SELECTION

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Optimum Laps is used to determine the final drive ratio by simulating the car's performance across a specific track layout, such as the Formula Bharat endurance or autocross circuits. The process begins by inputting the vehicle's "point mass" parameters, including the SAIETTA 119R motor's torque curve, vehicle mass, tire friction coefficients, and aerodynamic drag. Then multiple simulations are run- a Sensitivity Analysis - varying the final drive ratio (e.g., from 3.5:1 to 4.5:1) in small increments. Instead of picking a range at random, we use the competition's "physical boundaries" to define a logical range for our analysis. We choose a range depending on the top speed and maximum torque we want for our vehicle.

```
masstotal_car = 320; %% Vehicle + Driver Weight
g = 9.81; %% Value of g
cd = 0.65; %% Drag Coefficient
cr = 0.021; %% Surface Friction Coefficient
a = 0.887; %% Frontal Area
rho = 1.15; %% Air Density at Kari Track in Jan (Mean)

k1 = 0.5*rho*cd*a;

af = 0.00051; %% Coefficient of efficiency of speed on the force of rolling resistance
theta = 0; %% Gradient at Track
Rw = 0.3556; %% Radius of Wheel
G = 5.1; %% Gear Ratio
DT_eff = 0.9; %% Drivetrain Efficiency
M_eff = 0.85; %% Motor Efficiency
Ce = 0.9; %% Motor Controller Efficiency

k2 = cr*masstotal_car*g*cos(theta);
F_gr = masstotal_car*g*sin(theta);
```

Fig- Matlab data of SAIETTA 119R motor for Lap simulations.

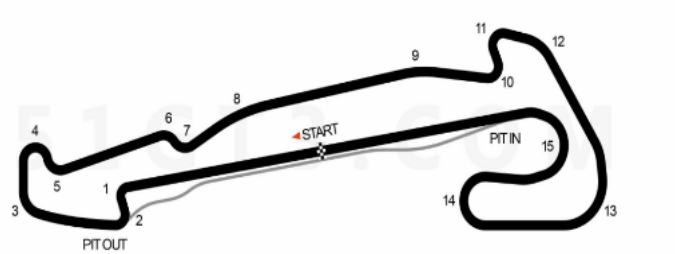

Fig. Kari Motor Speedway, Coimbatore, Tamil Nadu

From simulations, we found our optimal range as 4.9:1 to 5.7:1. We ran Optimum Lap for every increment (4.9, 5.4, 5.6...) within that 4.9 to 5.7 range for the Kari motor Speedway Track. In Formula Student, selecting a drive ratio based on Acceleration and Endurance is crucial as these two events represent the most extreme and most sustained performance requirements of the car. The optimization curves for two different events- Acceleration (This measures the time taken to cover a 75-meter straight line from a standing start) and Endurance (for reliability and energy efficiency) are provided by the software.

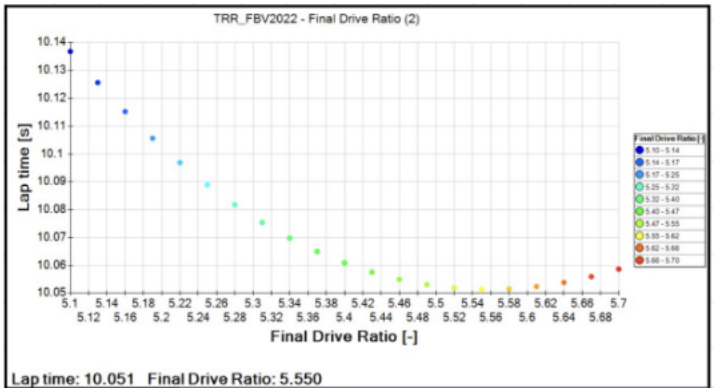

Fig 7. Acceleration event curve

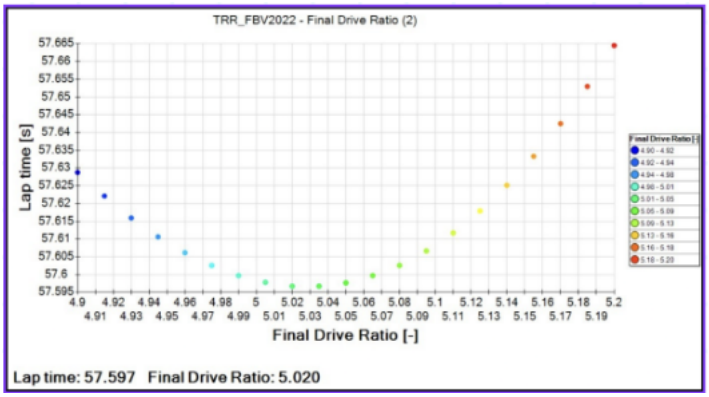

Fig 8. Endurance event curve

The final drive ratio is calculated as the average of the Acceleration and Endurance ratios, using the event points as weights:

$$\text{Final Ratio} = \frac{(\text{Ratio}_{\text{Accel}} \times \text{Points}_{\text{Accel}}) + (\text{Ratio}_{\text{Endurance}} \times \text{Points}_{\text{Endurance}})}{\text{Points}_{\text{Accel}} + \text{Points}_{\text{Endurance}}}$$

$$\text{Final Ratio} = \frac{(5.55 \times 75) + (5.02 \times 325)}{75 + 325}$$

$$\text{Final Ratio} = \frac{416.25 + 1631.50}{400}$$

$$\text{Final Ratio} = \frac{2047.75}{400}$$

$$\text{Precise Ratio} = \mathbf{5.119375}$$

The precise ratio of 5.119375 is rounded to 5.1 for manufacturing practicality, as the performance difference is negligible.

The core tradeoff: Acceleration vs Top Speed

A notable tradeoff noticed while optimizing the drive ratio of our vehicle was the speed-acceleration tradeoff. In gear systems, torque and speed are always in a trade-off relationship. At a fundamental level, increasing torque automatically reduces speed, and increasing speed reduces the available torque. We noticed the same for our chain drive transmission with sprockets. High Gear Ratio multiplies the torque from the motor which leads to rapid acceleration, but the engine hits its maximum RPM at a very low road speed. Low Gear Ratio reduces the torque multiplication but allows the wheels to spin faster relative to the engine RPM.